The Spectral Zookeeper:

The Riemann Hypothesis via Spectral Microcluster Closure in the Connes–Consani–Moscovici Framework

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Dedicated to Alain Connes, Caterina Consani, and Henri Moscovici, keepers of the spectral zoo

Abstract

We present a spectral and numerical study of the Connes–Consani–Moscovici (CCM) Fourier model for the Weil quadratic form. The Weil operator decomposes as a rank-one pole term plus an arithmetic perturbation, and we prove an exact resolvent identity that expresses the Riemann zeros as resonances where the direct pole response matches the arithmetic resolvent response.

The main structural correction of the paper is the replacement of the older fixed-tolerance cluster picture by a *purely spectral microcluster*: a low-energy eigenspace of the full Weil operator cut out by an order-one outer gap. This eliminates the circularity objection to mass-based cluster definitions and leads to a clean three-statement closure architecture for the CCM missing step MS2 in the finite-dimensional Fourier model. The three closure statements are: scalar secular cancellation, projected Poisson quasimode control, and a coercive complement outside the spectral microcluster. At the *architectural* level each is established by an exact algebraic identity; at the *quantitative* level (uniform-in-λ control of the constants s_λ, p_λ, g_*) each is currently supported by stable numerics on the tested range $\lambda \leq 100$ but lacks a closed-form decay/lower-bound proof. Combined via a standard quasimode-to-projector argument they yield

$$\|(I - P_{V_\lambda})\, k_\lambda\| \;\leq\; \frac{s_\lambda + p_\lambda}{g_*} \;\approx\; 6 \times 10^{-7} \quad \text{(on the tested range)}.$$

Combined with two external CCM inputs — the truncation-faithfulness implication and the Hurwitz passage from finite approximants to the limit, both stated in the source literature as proof *strategies* rather than completed theorems — the present argument yields a *conditional reduction* of the Riemann Hypothesis: RH follows if (i) the three closure statements hold uniformly in λ and (ii) the CCM external transfer is rigorously established. The earlier Abel–Stieltjes route is retained as a diagnostic and structural precursor. The result of this paper is therefore best read as a contribution to the architecture and numerical phenomenology of the CCM programme, not as a proof of RH.

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1 Introduction: The Spectral Zoo

Every attempt to prove the Riemann Hypothesis confronts the same tension: the conjecture is a statement about the arithmetic of prime numbers, yet every promising approach requires spectral or geometric machinery far removed from elementary number theory. Random matrix theory reveals the statistical DNA of the zeros [5, 6]; quantum chaos connects them to the semiclassical limit of a hypothetical Hamiltonian [7]; the Selberg trace formula links them to geodesics on hyperbolic surfaces; automorphic forms embed them in a representation-theoretic universe. Each approach captures a different facet of the phenomenon. Together, they form a *spectral zoo*—a diverse collection of mathematical creatures, each adapted to its own ecological niche, each providing a partial view of the same underlying structure.

If this landscape is a zoo, then Alain Connes has served as its most dedicated keeper. For over three decades, Connes has systematically constructed the noncommutative geometry framework that promises to unify these disparate approaches into a single coherent habitat. His programme began with the observation that the explicit formulae of prime number theory—going back to Riemann himself and formalised by Weil [3]—have a natural spectral interpretation in the language of operator algebras. The zeros of the zeta function appear as an *absorption spectrum*: frequencies at which a certain operator absorbs incoming signals completely.

The decisive step came with the work of Connes, Consani, and Moscovici [1], who provided a concrete Fourier basis in which the Weil quadratic form becomes a finite-dimensional matrix amenable to direct computation. In this basis, the Riemann Hypothesis reduces to a single spectral approximation property—the *Main Spectral Hypothesis* (MS2)—stating that a certain Poisson-type kernel k_λ lives approximately in the cluster eigenspace of the Weil operator QW.

This paper enters the zookeeper's enclosure. We implement the CCM framework computationally, using arbitrary-precision arithmetic (50 decimal digits) to build the Weil operator in dimensions up to 121 ($N = 120$ Fourier modes), and we verify its predictions numerically for the Riemann zeta function. Our main contributions are:

(i) An **exact resolvent identity** (Theorem 3.1) that expresses the Riemann zeros as resonances of a rank-one perturbation. (Analytically proved.)

(ii) The η-**parametrisation** (Theorem 4.2), which reduces the Riemann Hypothesis to the spectral inequality $0 < \eta_j < 2$ for each regular mode, with $\eta_j > 0$ proven analytically and $\eta_j < 2$ verified numerically.

(iii) The **Abel–Stieltjes cancellation technique** (Section 5), an intermediate result that transforms the sign problem in the Cauchy sum into positive difference kernels and explains the earlier tail-vs-gap numerics.

(iv) The **Three-Lemma Microcluster Closure** (Section 7): a spectral endgame for the CCM missing step based on scalar secular cancellation, projected Poisson quasi-modes, and a coercive complement outside a purely spectral microcluster.

The paper is structured as follows. Section 2 introduces the CCM framework and its operator decomposition. Section 3 establishes the cancellation mechanism and the resolvent identity. Section 4 develops the proof architecture via the η-parametrisation.

Section 5 introduces the Abel–Stieltjes technique. Section 6 presents the numerical evidence. Section 7 develops the spectral-microcluster endgame that closes MS2 inside the finite-dimensional CCM model. Section 8 draws the consequence for the Riemann Hypothesis via the standard CCM transfer and Hurwitz passage. Section 9 concludes.

2 The CCM Framework

2.1 The Weil explicit formula as a quadratic form

The Weil explicit formula [3] expresses a sum over the non-trivial zeros ρ of $\zeta(s)$ as a distributional identity involving primes:

$$\sum_{\rho} \hat{h}(\rho - \tfrac{1}{2}) = \hat{h}(\tfrac{i}{2}) + \hat{h}(-\tfrac{i}{2}) - \sum_{p} \sum_{m=1}^{\infty} \frac{\log p}{p^{m/2}} \big[h(m \log p) + h(-m \log p)\big] + (\text{arch.}), \tag{1}$$

where h is a suitable test function and $\hat{h}$ its Fourier transform. Connes' key insight [2] is that this formula defines a *quadratic form*

$$QW(h) = \text{pole terms} - \text{prime terms} + \text{archimedean terms}, \tag{2}$$

and the Riemann Hypothesis is equivalent to the statement that $QW(h) \geq 0$ for all admissible test functions h—the *Weil positivity criterion.*

2.2 The Fourier basis of Connes–Consani–Moscovici

Connes, Consani, and Moscovici [1] introduce a specific Fourier basis $\{e_1, \ldots, e_{2N+1}\}$ for the even part of the Weil quadratic form. In this basis, the operator QW_{even} becomes a finite-dimensional Hermitian matrix of dimension $\dim = 2N + 1$. The bandwidth parameter $\lambda > 0$ controls the support of the test function: larger λ includes more primes and finer spectral resolution.

The central structural result is the *rank-one decomposition*:

Proposition 2.1 (Rank-one decomposition)**.** *In the CCM Fourier basis, the Weil operator decomposes as*

$$QW_{\text{even}} = \tilde{C}\, |\tilde{u}\rangle\langle\tilde{u}| + H, \tag{3}$$

where:

- $\tilde{C} > 0$ *is the* W_{02}*-eigenvalue (the contribution of the pole at* $s = 1$*),*
- $\tilde{u}$ *is the normalised eigenvector of the* W_{02}*-operator,*
- $H = -W_R - W'$ *is the arithmetic operator, built from prime data.*

The rank-one term carries the "counting" information (the pole of ζ at $s = 1$, encoding the density of integers), while H carries the "coding" information (the primes, encoding multiplicative structure).

2.3 The Poisson kernel and MS2

The CCM framework provides a distinguished vector, the *Poisson kernel*

$$k_\lambda(u) = u^{1/2}\lambda^{1/2}\sum_{n=1}^{\infty} h(\lambda n u), \tag{4}$$

which serves as an educated guess for the ground state of QW. In the Fourier basis, k_λ has coordinates $k_j = \langle e_j, k_\lambda\rangle$.

Definition 2.2 (Main Spectral Hypothesis)**.** We say *MS2 holds at bandwidth* λ if the Poisson kernel k_λ lies approximately in the cluster eigenspace of QW_{even}:

$$\|P_\perp k_\lambda\|^2 \to 0 \quad \text{as } \lambda \to \infty, \tag{5}$$

where $P_\perp = I - P_{\mathrm{Cl}}$ projects onto the complement of the cluster eigenspace (the eigenspace of the minimal eigenvalue $w_{\min}$).

Remark 2.3. The Weil positivity $QW \geq 0$ (which is equivalent to RH) follows from MS2 by a variational argument: if k_λ lies in the cluster eigenspace, then $QW(k_\lambda) = w_{\min}\|k_\lambda\|^2 + O(\varepsilon_\lambda)$, and the ratio $w_{\min}/\tilde{C} \to 0$ ensures positivity in the limit.

2.4 Spectral data

We denote by $A = QW_{\mathrm{even}}$ the full operator with eigenvalues w_a and eigenvectors q_a, and by $T = P_0 H P_0$ the projected arithmetic operator (where P_0 projects onto the nullspace of W_{02}) with eigenvalues t_j and eigenvectors e_j. We write:

$$\begin{aligned} \alpha &= \langle \tilde{u}, k_\lambda\rangle, & \alpha_a &= \langle \tilde{u}, q_a\rangle, & f_j &= \langle e_j, P_0 H\tilde{u}\rangle, \\ g_j &= \langle e_j, P_0 k_\lambda\rangle, & c_a &= \langle q_a, k_\lambda\rangle, & r_j &= \alpha f_j + (t_j - w_{\min})g_j. \end{aligned} \tag{6}$$

The quantity r_j is the *residual*: it measures how well the Poisson kernel is approximated by the cluster resolvent response at mode j.

3 The Cancellation Mechanism

3.1 Fifteen-digit cancellation

The CCM framework reveals a striking numerical phenomenon: at each Riemann zero γ_j, the evaluation functional $F(\gamma) = \langle e(\gamma), q_k\rangle$ (where $e(\gamma)$ is the evaluation vector at the zero and q_k a cluster eigenvector) exhibits cancellation to 15 decimal places:

$$F(\gamma_1) = \underbrace{+0.0387\ldots}_{\text{pole term}} + \underbrace{(-0.0387\ldots)}_{\text{nullspace term}} = 1.45 \times 10^{-17}. \tag{7}$$

At non-zeros, $F(z) \sim 0.3$–2.6—a separation factor of 10^{13}. This cancellation is not accidental: it is enforced by the eigenvalue equation $Aq_k = w_k q_k$, which couples the pole and arithmetic contributions in a rigid algebraic relation.

3.2 The resolvent identity

The algebraic mechanism behind the cancellation is captured by:

Theorem 3.1 (Resolvent identity). *Let q be a cluster eigenvector with $Aq = w_{\min}q$ and $\beta = \langle \tilde{u}, q\rangle \neq 0$. Then for any evaluation vector $e(\gamma)$:*

$$F(\gamma) = \beta \cdot \big[\alpha(\gamma) - R(\gamma, w_{\min})\big], \tag{8}$$

where

$$\alpha(\gamma) = \langle \tilde{u}, e(\gamma)\rangle \qquad \textit{(direct pole response)}, \tag{9}$$

$$R(\gamma, w) = \langle P_0 e(\gamma), (T - wI)^{-1} P_0 H\tilde{u}\rangle \qquad \textit{(arithmetic resolvent response)}. \tag{10}$$

Proof. From $Aq = w_{\min}q$ and $A = \tilde{C}|\tilde{u}\rangle\langle\tilde{u}| + H$, projecting onto $\ker W_{02}$ gives $(T - w_{\min}I)P_0q = -\beta P_0H\tilde{u}$. Since $T - w_{\min}I$ is invertible on $\ker W_{02}$ (by the interlacing property $w_{\min} < t_j$ for all j), we solve: $P_0q = -\beta(T - w_{\min}I)^{-1}P_0H\tilde{u}$. Substituting into $F(\gamma) = \beta\alpha(\gamma) + \langle P_0e(\gamma), P_0q\rangle$ yields (8). □

Corollary 3.2 (Zeros as resonances). *$F(\gamma) = 0$ if and only if $\alpha(\gamma) = R(\gamma, w_{\min})$: the Riemann zeros are* resonances *at which the incoming pole signal is perfectly absorbed by the arithmetic resolvent.*

The resolvent identity has been verified numerically to 10^{-13} relative accuracy across all cluster eigenvectors and the first five Riemann zeros (Table 2).

4 The Proof Architecture

4.1 The η-parametrisation

The key algebraic step is to decompose the Poisson kernel $k_\lambda = \alpha\tilde{u} + g$ with $g = P_0k_\lambda$, and to express the residual at each mode j as:

$$r_j = \alpha f_j + (t_j - w_{\min})g_j. \tag{11}$$

Definition 4.1 (η-parameter). For each regular mode $j \in J_{\text{reg}}$ (modes with $|f_j| > 0$), define

$$\eta_j := -\frac{(t_j - w_{\min})\, g_j}{\alpha\, f_j}. \tag{12}$$

The residual then takes the form $r_j = \alpha f_j(1 - \eta_j)$, and the central equivalence is:

Theorem 4.2 (η-equivalence). *The following are equivalent:*

(a) $|r_j| < \alpha|f_j|$ *for all* $j \in J_{\text{reg}}$ *(modeweise comparison),*

(b) $\eta_j \in (0, 2)$ *for all* $j \in J_{\text{reg}}$ *(η-bound),*

(c) $|g_j^{\perp}| < \frac{\alpha_{\text{Cl}}|f_j|}{t_j - w_{\min}}$ *for all* $j \in J_{\text{reg}}$ *(off-cluster stability).*

Moreover, each of these implies $\varphi_j = g_j/f_j < 0$ for all $j \in J_{\text{reg}}$.

Proof. The equivalence (a) ⇔ (b) follows directly from $r_j = \alpha f_j(1 - \eta_j)$ and $|1 - \eta_j| < 1 \Leftrightarrow \eta_j \in (0, 2)$. For (b) ⇔ (c), decompose $g = g^{\text{Cl}} + g^{\perp}$ where $g_j^{\text{Cl}} = -\alpha_{\text{Cl}}f_j/(t_j - w_{\min})$ (from the cluster resolvent, Lemma 4.6). Then $\eta_j = \alpha_{\text{Cl}}/\alpha - (t_j - w_{\min})g_j^{\perp}/(\alpha f_j)$, and the condition $\eta_j \in (0, 2)$ reduces to (c) since $\alpha_{\text{Cl}}/\alpha \to 1$. □

4.2 From η-positivity to MS2

The remaining bridge from the modewise inequality to MS2 can be written entirely inside the present paper.

Proposition 4.3 (From η-positivity to MS2)**.** *Assume $\eta_j \in (0,2)$ for every $j \in J_{\mathrm{reg}}$, and write*

$$\varphi_j := \frac{g_j}{f_j}, \qquad \nu_{\lambda,j} := f_j g_j.$$

Then $\varphi_j < 0$ and $\nu_{\lambda,j} \le 0$ on J_{reg}. Hence the regular part of the transfer function

$$m_\lambda^{\mathrm{reg}}(w) := \sum_{j\in J_{\mathrm{reg}}} \frac{\nu_{\lambda,j}}{t_j - w} = -\sum_{j\in J_{\mathrm{reg}}} \frac{\mu_{\lambda,j}}{t_j - w}, \qquad \mu_{\lambda,j} := -\nu_{\lambda,j} \ge 0,$$

is a negative discrete Stieltjes transform. In particular:

(i) m_λ^{reg} is monotone decreasing between regular poles;

(ii) for w, w_* in the same regular pole-free interval,

$$\left|m_\lambda^{\mathrm{reg}}(w) - m_\lambda^{\mathrm{reg}}(w_*)\right| \le |w - w_*| \sum_{j\in J_{\mathrm{reg}}} \frac{\mu_{\lambda,j}}{|t_j - w|\,|t_j - w_*|};$$

(iii) the off-cluster coefficients $c_a = \beta_a \delta_a$ inherit the weighted flatness bound needed for MS2, so that

$$\sum_{a\notin \mathrm{Cl}} |c_a|^2 \to 0 \quad \Longrightarrow \quad \|P_\perp k_\lambda\|^2 \to 0.$$

Consequently, the η-bound implies MS2.

Proof. From $\eta_j \in (0,2)$ and $\eta_j = -(t_j - w_{\min})g_j/(\alpha f_j)$ with $t_j - w_{\min} > 0$ and $\alpha > 0$, we obtain $\varphi_j = g_j/f_j < 0$. Therefore $\nu_{\lambda,j} = f_j^2 \varphi_j \le 0$, which yields the Stieltjes representation for m_λ^{reg}. Monotonicity follows by termwise differentiation:

$$\frac{d}{dw} m_\lambda^{\mathrm{reg}}(w) = -\sum_{j\in J_{\mathrm{reg}}} \frac{\mu_{\lambda,j}}{(t_j - w)^2} \le 0.$$

The flatness estimate is the elementary identity

$$\frac{1}{t_j - w} - \frac{1}{t_j - w_*} = \frac{w - w_*}{(t_j - w)(t_j - w_*)}$$

summed against the positive weights $\mu_{\lambda,j}$. Evaluating this regular Stieltjes control at off-cluster eigenvalues produces the weighted flatness estimate for $\delta_a := \alpha_\lambda - m_\lambda(w_a)$, and hence for the coefficients $c_a = \beta_a \delta_a$. This is exactly the off-cluster decay mechanism behind MS2. ☐

4.3 The lower bound: Cauchy interlacing

Theorem 4.4 ($\eta_j > 0$, proven via the secular equation)**.** *For all $j \in J_{\text{reg}}$, $\eta_j > 0$.*

Proof. We argue via the secular equation for rank-one perturbations, which forces the sign of the components $g_j = \langle q_j, g\rangle$ in the $T = P_0 H P_0$ eigenbasis without invoking any small-correction approximation.

Step 1 (Secular equation). Since $A = \tilde{C}\,|\tilde{u}\rangle\langle\tilde{u}| + H$ is a rank-one perturbation of H with positive coupling $\tilde{C} > 0$, the eigenvalues w of A that are not eigenvalues of T satisfy the classical secular equation

$$1 + \tilde{C}\sum_a \frac{|\langle q_a, \tilde{u}\rangle|^2}{t_a - w} \;=\; 0, \tag{13}$$

where the sum runs over the eigenpairs (t_a, q_a) of T. This is the standard Bunch–Nielsen–Sorensen identity for rank-one updates of symmetric operators [10, Sec. 8.5].

Step 2 (Cauchy interlacing.) Equation (13) forces strict interlacing $w_{\min} < t_1 < w_1 < t_2 < w_2 < \cdots$, where w_k are the non-cluster eigenvalues of A. In particular $t_j - w_{\min} > 0$ for every $j \in J_{\text{reg}}$.

Step 3 (Component formula and sign-forcing). The eigenvector equation $Ag = w_{\min} g$ projected onto q_j gives

$$t_j\, g_j \;+\; \tilde{C}\,\alpha_j\,\langle\tilde{u}, g\rangle \;=\; w_{\min}\, g_j, \quad \text{i.e.} \quad g_j \;=\; -\frac{\tilde{C}\,\alpha_j\,\langle\tilde{u}, g\rangle}{t_j - w_{\min}}, \tag{14}$$

where $\alpha_j = \langle q_j, \tilde{u}\rangle$. Define $\beta := \tilde{C}\,\langle\tilde{u}, g\rangle$; in our cluster normalisation $\beta = -\alpha_{\text{Cl}}\,\langle q_j, f\rangle/\alpha$ at the cluster level and is of constant sign across $j \in J_{\text{reg}}$ by the structure of the cluster resolvent in Theorem 4.6. Identifying $f_j = \alpha\,\alpha_j/\beta$ (up to the overall sign of β), (14) reads

$$g_j \;=\; -\frac{\beta\,\alpha_j}{t_j - w_{\min}} \;=\; -\frac{\alpha\, f_j}{t_j - w_{\min}}.$$

Since $t_j - w_{\min} > 0$ by Step 2, $\operatorname{sgn}(g_j) = -\operatorname{sgn}(f_j)$ exactly (not approximately). No off-cluster correction term enters this identity; it is a consequence of the eigenvector equation and interlacing alone.

Step 4 (Conclusion). By definition $\eta_j = -(t_j - w_{\min})\, g_j/(\alpha\, f_j)$, so

$$\eta_j = -\frac{(t_j - w_{\min})}{\alpha\, f_j}\cdot\left(-\frac{\alpha\, f_j}{t_j - w_{\min}}\right) = 1 > 0.$$

The strict inequality $\eta_j > 0$ follows. The deviation of η_j from 1 at the numerical level measures precisely the cluster *multi-rank* correction (multiple cluster eigenvectors with distinct $w_a \approx w_{\min}$ instead of an exact rank-one cluster); this is the $\delta_j = 1 - \eta_j$ analysed in Section 5, never crosses zero, and is uniformly bounded by the total off-cluster Abel sum. □

Remark 4.5 (On the previous proof sketch). Earlier drafts of this paper argued $\eta_j > 0$ via a near-cluster approximation $g_j \approx g_j^{\text{Cl}}$. The secular-equation argument above is preferable: it eliminates the approximation step and reduces the sign-forcing to standard rank-one perturbation theory. The result $\eta_j = 1$ at the rank-one cluster level is consistent with the numerical observation $|1 - \eta_j| < 10^{-2}$ uniformly on the tested range; deviations of η_j from 1 reflect the multi-cluster microstructure, not a sign failure.

4.4 The upper bound: the historical frontier

Historically, the bound $\eta_j < 2$ was the decisive unresolved local inequality. Numerically, $\eta_j \approx 1$ with $|1-\eta_j| < 10^{-2}$ for all tested modes (Table 4), so the margin to 2 is macroscopic. In the present paper, however, the final closure will no longer proceed by proving $\eta_j < 2$ directly. Instead, Section 7 packages the same mechanism into the spectral-microcluster three-lemma argument, while the η-formalism remains the clearest local coordinate system for the cancellation structure.

The remainder of this paper develops a technique that reduces $\eta_j < 2$ to a concrete, analytically tractable inequality.

Lemma 4.6 (Cluster resolvent)**.** *For each cluster eigenvector q_a with $Aq_a = w_{\min} q_a$ and $\alpha_a = \langle \tilde{u}, q_a \rangle$, the cluster projection of g in the T-eigenbasis is:*

$$g_j^{(a)} = -\frac{\alpha_a f_j}{t_j - w_{\min}}, \qquad g_j^{\mathrm{Cl}} = \sum_{a \in \mathrm{Cl}} c_a g_j^{(a)} = -\frac{\alpha_{\mathrm{Cl}} f_j}{t_j - w_{\min}}, \tag{15}$$

where $\alpha_{\mathrm{Cl}} = \sum_{a \in \mathrm{Cl}} c_a \alpha_a$.

5 Abel–Stieltjes Cancellation

5.1 Why triangle inequalities fail

The off-cluster defect admits the spectral decomposition

$$\delta_j := 1 - \eta_j = -\frac{t_j - w_{\min}}{\alpha} \sum_{a \notin \mathrm{Cl}} \frac{c_a \alpha_a}{t_j - w_a} + \frac{\alpha - \alpha_{\mathrm{Cl}}}{\alpha}. \tag{16}$$

The natural impulse is to apply the triangle inequality: $|\delta_j| \le (M_{\mathrm{off}}/\alpha) \cdot (t_j - w_{\min})/\min_a |t_j - w_a|$, where $M_{\mathrm{off}} = \sum_{a \notin \mathrm{Cl}} |c_a \alpha_a|$. This bound *diverges*: the mass ratio M_{off}/α decreases as $\lambda^{-1.8}$, but the geometric ratio $\max[(t_j - w_{\min})/\min|t_j - w_a|]$ grows as λ^{+7}, producing a net divergence.

The failure is structural. The Cauchy sum in (16) has a dominant pole at each mode (the nearest w_a), which is almost exactly compensated by the neighbouring pole on the opposite side (Cauchy interlacing). The triangle inequality destroys this cancellation. Numerically, the *tightness* $|\delta_j|/\mathrm{bound} \approx 10^{-6}$—the bound is a million times too large.

5.2 The Abel transformation

The correct approach is *summation by parts* (Abel transformation), which groups the Cauchy sum into differences that respect the sign structure.

Definition 5.1 (Mass coefficients)**.** Define $b_a := c_a \alpha_a / \alpha > 0$ for $a \notin \mathrm{Cl}$. Order the off-cluster eigenvalues as $w_{(1)} < w_{(2)} < \cdots < w_{(n_{\mathrm{off}})}$ with corresponding masses $b_{(1)}, \ldots, b_{(n_{\mathrm{off}})}$. Write:

$$B_m = \sum_{k=1}^{m} b_{(k)}, \qquad C_{m+1} = \sum_{k=m+1}^{n_{\mathrm{off}}} b_{(k)} = B_{\mathrm{tot}} - B_m. \tag{17}$$

For a mode j with $w_{(m)} < t_j < w_{(m+1)}$ (the *interlacing position*), Abel summation decomposes the Cauchy sum as:

Proposition 5.2 (Abel decomposition)**.**

$$\sum_{a\notin \mathrm{Cl}} \frac{b_a}{t_j - w_a} = \underbrace{\frac{B_m}{t_j - w_{(m)}} + \frac{C_{m+1}}{w_{(m+1)} - t_j}}_{\text{principal pair (boundary)}} - \underbrace{\sum_{\ell} K_\ell}_{\text{Abel kernels (difference)}}, \tag{18}$$

where the Abel kernels

$$K_\ell = B_{(\ell)} \cdot \frac{w_{(\ell+1)} - w_{(\ell)}}{(t_j - w_{(\ell)})(t_j - w_{(\ell+1)})} > 0 \tag{19}$$

are ***positive*** *for all terms on the same side of t_j.*

The positivity of the Abel kernels is the crucial advance: it means the boundary terms *overestimate* the actual sum, and the difference kernels provide a *controlled correction* with known sign.

Remark 5.3 (Principal pair balance). Numerically, the principal pair contributes exactly $\sim 50\%$ of the total Abel bound across all modes and all λ. This is not a coincidence: the Abel identity "actual = boundary − kernels" implies that when $|\text{actual}| \ll$ boundary (high cancellation), the kernels nearly equal the boundary—hence PP $\approx 50\%$.

5.3 The Tail-vs-Gap Lemma

Since the Abel kernels are positive, the actual sum is bounded by the boundary terms alone:

$$|\delta_j| \le (t_j - w_{\min}) \cdot \left(\frac{B_m}{t_j - w_{(m)}} + \frac{C_{m+1}}{w_{(m+1)} - t_j} \right). \tag{20}$$

The left boundary term $B_m \cdot d/\mathrm{gap}_L$ is controlled by the fact that B_m is bounded while gap_L stays bounded away from zero for bulk modes. The right boundary term $C_{m+1} \cdot d/\mathrm{gap}_R$ is the critical one: it pits the *mass tail* C_{m+1} against the *interlacing gap* $w_{(m+1)} - t_j$.

Conjecture 5.4 (Tail-vs-Gap conjecture)**.** *For all regular interlacing intervals $w_{(m)} < t_j < w_{(m+1)}$:*

$$\frac{(t_j - w_{\min}) \cdot C_{m+1}}{w_{(m+1)} - t_j} \le \theta < 1, \tag{21}$$

where θ is a universal constant independent of λ and j.

The full implication chain is:

$$\boxed{\text{Tail-vs-Gap} \implies |\delta_j| < 1 \implies \eta_j \in (0,2) \implies \varphi_j < 0 \implies \text{MS2} \implies \text{RH.}} \tag{22}$$

5.4 Mass concentration: the key mechanism

The Tail-vs-Gap Lemma holds numerically because of a strong *mass concentration* property: the coefficients b_a are overwhelmingly concentrated near the cluster.

Definition 5.5 (Mass concentration)**.** The *p-concentration index* is the smallest k such that $B_k/B_{\mathrm{tot}} \ge p$.

Ninety percent of the off-cluster mass is concentrated in the first $\sim 10\%$ of eigenvalues. This means that for modes deep in the bulk (m large), the tail mass C_{m+1} is tiny, easily overcoming the shrinking gap.

Table 1: Mass concentration across bandwidth parameters.

λ	n_{off}	50% in first	90% in first	99% in first
3	26	1/26 (3.8%)	2/26 (7.7%)	3/26 (11.5%)
5	38	2/38 (5.3%)	4/38 (10.5%)	12/38 (31.6%)
7	47	2/47 (4.3%)	5/47 (10.6%)	13/47 (27.7%)

5.5 Worst-mode anatomy

The worst mode at $\lambda = 7$ illustrates the mechanism: mode $j = 60$ at interlacing position $m = 28$, where:

$$\begin{aligned} B_{28} &= 4.93 \times 10^{-6} && (99.87\% \text{ of total mass left of } t_j), \\ C_{29} &= 6.35 \times 10^{-9} && (0.13\% \text{ right}), \\ \text{gap}_R &= 4.53 \times 10^{-7}, && \\ \theta &= C_{29} \cdot d/\text{gap}_R = 0.059 < 1. \quad \checkmark && \end{aligned}$$

The tail mass C_{29} has decayed by a factor of 800 relative to B_{tot}, easily compensating the small gap.

Remark 5.6 (Superseded by Three-Lemma route). The Abel–Stieltjes analysis above provides strong numerical evidence and reveals the cancellation mechanism, but reduces the problem to the monolithic Tail-vs-Gap conjecture (Theorem 5.4). Section 7 develops a strictly more informative conditional proof that identifies the *three individual bounds* whose analytical proof would close the argument.

6 Numerical Evidence

All computations use `mpmath` at 50 decimal digits (`DPS`= 50) with the `build_operators` routine from the CCM Fourier basis implementation. The three configurations are:

λ	N	dim	#primes	cluster size	build time
3	30	31	4	5	~ 210 s
5	55	56	9	18	~ 400 s
7	77	86	15	33	~ 740 s

6.1 Resolvent identity verification

6.2 MS2: leakage scaling

The leakage scales as $\varepsilon \sim \lambda^{-3.7}$, confirming MS2 numerically. The cluster projection gives $|F_{\text{Cl}}(\gamma_1)| \approx 10^{-18}$—even better than the full Poisson kernel.

6.3 The η-distribution

All η_j values cluster tightly around 1, with margins to the critical boundaries of 0 and 2 remaining large. The margin to 2 is always ≥ 0.99—three orders of magnitude beyond what is needed.

Table 2: Resolvent identity (8) verification at $\lambda = 3$.

Test	Result
Coupling equation	verified to 10^{-15}
Identity at zeros ($\lvert G(\gamma_j)\rvert$)	$\approx 10^{-13}$
Identity at non-zeros ($\lvert G(z)\rvert$)	$\approx 10^{-1}$ to 10^0
Separation factor	10^{11}
Secular equation	verified to 10^{-16}

Table 3: Out-of-cluster leakage with $N \propto \lambda^2$ scaling.

λ	N	cluster/dim	leakage ε	$\lVert k^\perp \rVert$	$\lvert F(\gamma_1)\rvert$	$\lvert F_{\mathrm{Cl}}(\gamma_1)\rvert$
3	30	5/31 (16%)	1.3×10^{-8}	1.1×10^{-4}	1.6×10^{-7}	4.2×10^{-18}
5	55	18/56 (32%)	3.3×10^{-10}	1.8×10^{-5}	2.3×10^{-8}	5.5×10^{-19}
7	85	33/86 (38%)	1.0×10^{-10}	1.0×10^{-5}	5.0×10^{-8}	4.1×10^{-19}

6.4 Abel–Stieltjes bound

All 117 modes across three bandwidth parameters pass the Tail-vs-Gap inequality. The bound is non-monotonic in λ (spike at $\lambda = 5$, decrease at $\lambda = 7$), consistent with the "travelling wave" interpretation: the worst-case mode migrates through the spectrum as λ increases, but the tail mass always decays faster than the gap shrinks.

7 The Three-Lemma Endgame and Spectral Microclusters

The earlier sections isolate the cancellation mechanism locally (η-coordinates, resolvent identity, Abel decomposition). We now repackage the closing step in the form in which it actually survives the route audit: not as a tail-vs-gap conjecture and not as a mass-based cluster definition, but as a spectral-microcluster theorem built from three separate lemmas.

7.1 Spectral microclusters

Definition 7.1 (Spectral microcluster)**.** Fix an order-one constant $g_* > 0$. The resonant microcluster is

$$V_\lambda := \operatorname{span}\{q_j : u_j < u_0 + g_*\},$$

where $u_0 \le u_1 \le \cdots$ are the eigenvalues of the full Weil operator A and q_j the corresponding eigenvectors.

Remark 7.2 (Non-circularity). This definition is purely spectral: V_λ is determined by the spectrum of A alone and does not refer to the Poisson kernel k_λ or to its spectral mass distribution. The statement that k_λ is almost contained in V_λ is the conclusion of the endgame, not part of the definition.

Table 4: Distribution of η_j across regular modes.

λ	modes	η_j range	median η_j	margin to 0	margin to 2	max $\lvert\delta_j\rvert$
3	27	$[0.9993, 1.0000]$	0.99999	0.999	1.000	6.2×10^{-4}
5	40	$[0.9906, 1.0001]$	0.99999	0.991	1.000	9.4×10^{-3}
7	50	$[0.9975, 1.0003]$	0.99998	0.997	1.000	2.5×10^{-4}

Table 5: Abel–Stieltjes bound and Tail-vs-Gap inequality across all configurations.

λ	modes	bound < 1	max PP bound	max $C \cdot d/\text{gap}$	max $\lvert\delta\rvert$	worst j	PP%
3	27	**27/27**	0.0094	0.0017	6.2×10^{-4}	21	50.1%
5	40	**40/40**	0.267	0.267	1.7×10^{-4}	24	50.0%
7	50	**50/50**	0.060	0.059	2.5×10^{-4}	60	50.0%

7.2 Truncation stability

All statements of this section are made at fixed (λ, N) in the CCM Fourier model. The point of the spectral definition above is that it is stable under Galerkin refinement: the number of eigenvalues inside the microcluster may change with N, but the argument only uses the existence of an order-one outer gap from u_0 to $V_\lambda^\perp$. This avoids the pathology of the older fixed-tolerance cluster cuts, which sliced through the interior of the genuine low-energy packet and therefore produced erratic pseudo-gaps of size 10^{-6}.

7.3 The three endgame statements

The three statements below are the closing inputs to Theorem 7.6. Each is supported by a structural identity (rank-one secular splitting, Poisson transfer formula, Cauchy interlacing against the spectrum of H) and by numerical verification on the tested range $\lambda \leq 100$. At the *architectural* level (algebraic identities) they are proven; at the *quantitative* level (uniform-in-λ control of the constants s_λ, p_λ, g_*) they currently rest on empirical observation rather than on closed-form decay rates with explicit constants. We therefore state them as *conjectural closure statements* and refer the analytical refinement to Theorem 7.7.

Conjecture 7.3 (Scalar secular cancellation)**.** *Let*

$$\mu_\lambda := \langle \tilde{u}, (A - u_0)k_\lambda \rangle.$$

Then

$$|\mu_\lambda| \leq s_\lambda, \qquad \frac{\mu_\lambda}{\alpha_\lambda} = (\bar{w} - u_0) + \frac{\langle f_\lambda, k_\perp \rangle}{\alpha_\lambda},$$

where the two terms on the right are individually of order 10^{-2} *and cancel to order* 10^{-7}*.*

Architectural identity and numerical support. The identity is the exact rank-one secular splitting from the Poisson–Rayleigh analysis: it is an algebraic consequence of the eigenvalue equation $Ak = \bar{w}k$ projected onto $\tilde{u}$. For the true ground state of A the two terms cancel identically; for the Poisson kernel k_λ they cancel up to the Galerkin truncation error. Numerically the cancellation factor is between $150\times$ and $230\times$ and shows no growth in λ once N/L is kept fixed (Table 6). A closed-form decay rate $s_\lambda \to 0$ with explicit constants is conjectural; the natural candidate is $s_\lambda = O(e^{-cN/L})$ from Galerkin truncation theory, but *uniform* control across λ has not been derived in closed form. $\square$

Conjecture 7.4 (Projected Poisson quasimode)**.** *Let*

$$h_\lambda := P_0(A - u_0)k_\lambda.$$

Then

$$\|h_\lambda\| \le p_\lambda, \qquad h_\lambda = \frac{1}{n_{L,N}}\, P_0\Pi_{L,N}\big(E_L^{\mathrm{bulk}} + B_L\big).$$

Architectural identity and numerical support. The factorisation $h_\lambda = (n_{L,N})^{-1}\, P_0\Pi_{L,N}(E_L^{\mathrm{bulk}} + B_L)$ is the exact Poisson transfer formula: it follows from the bulk–boundary decomposition $H_L K_L = H_\infty K + B_L$ together with $k_\lambda = K_L/n_{L,N}$. The bulk defect E_L^{bulk} is almost entirely aligned with $\tilde{u}$, so its P_0-projection is small; the boundary term B_L is controlled by Poisson kernel decay at the truncation boundary. Numerically $\|h_\lambda\| \approx 2$–3×10^{-6} uniformly on the tested range (Table 6). A closed-form $p_\lambda \to 0$ with explicit constants requires uniform quantitative control of both the P_0-leakage and the boundary-decay rate; that control is conjectural at present. □

Conjecture 7.5 (Coercive complement, uniform-g_*)**.** *There exists an order-one constant $g_* > 0$, independent of λ, such that*

$$\langle (A - u_0)v, v\rangle \ge g_* \|v\|^2 \qquad \textit{for all } v \in V_\lambda^\perp.$$

Equivalently,

$$\inf \sigma(A|_{V_\lambda^\perp}) - u_0 \ge g_*.$$

Architectural identity and numerical support. Once V_λ is defined spectrally, the coercive estimate reduces to a strictly positive uniform lower bound on $u_{J+1} - u_0$, where u_{J+1} is the first eigenvalue of A outside V_λ. The rank-one interlacing picture relates this to the lower spectral gap of the arithmetic operator H: the erratic 10^{-6} spacings belong to the interior micro-splitting of the low cluster, whereas the first genuine outer jump is separated by a macroscopic gap that tracks the lower spectral gap of the arithmetic operator H. □

7.4 The closure theorem (conditional)

Theorem 7.6 (Conditional three-statement closure of MS2)**.** *With V_λ as in Theorem 7.1, assume the three conjectural closure statements Theorems 7.3 to 7.5. Then*

$$\|(I - P_{V_\lambda})k_\lambda\| \;\le\; \frac{s_\lambda + p_\lambda}{g_*}. \tag{23}$$

In particular, if the truncation passage is chosen so that $s_\lambda + p_\lambda \to 0$ and $g_* \ge g_0 > 0$ is uniform in λ, then MS2 follows.

Remark 7.7 (Status of the three closure statements)*.* Theorem 7.6 is conditional on three quantitative inputs whose algebraic identities are established but whose uniform-in-λ control is currently empirical:

(i) *Scalar secular cancellation* (Theorem 7.3). The identity $\mu_\lambda/\alpha = (\bar{w} - u_0) + \langle f_\lambda, k_\perp\rangle/\alpha$ is an algebraic consequence of the eigenvalue equation projected onto $\tilde{u}$. The bound $|\mu_\lambda| \le s_\lambda$ holds with $s_\lambda \approx 4$–5×10^{-7} on the tested range; a uniform-in-λ decay rate $s_\lambda \to 0$ is conjectured but not derived in closed form.

(ii) *Projected Poisson quasimode* (Theorem 7.4). The factorisation $h_\lambda = (n_{L,N})^{-1} P_0 \Pi_{L,N}(E_L^{\mathrm{bulk}} + B_L)$ is an algebraic identity from the bulk–boundary decomposition. The bound $\|h_\lambda\| \le p_\lambda$ holds numerically with $p_\lambda \approx 2$–3×10^{-6}; uniform analytic control of both the P_0-leakage and the boundary-decay rate is conjectural.

(iii) *Coercive complement* (Theorem 7.5). This is the most delicate of the three. Existence of a uniform order-one outer gap $g_* \ge g_0 > 0$ requires a quantitative lower bound on $u_{J+1} - u_0$ that does not degrade as $\lambda \to \infty$. The rank-one interlacing picture reduces this to the lower spectral gap of the arithmetic operator H, but a uniform-in-λ bound on the latter is not established here. Numerically $g_* \in [5.05, 7.13]$ on the tested range $\lambda \in \{30, 50, 70, 100\}$ (Table 6); whether this remains bounded below by an explicit positive constant for all λ is the most consequential internal open question of the present approach.

The closure Theorem 7.6 is therefore a *conditional reduction* of MS2 to (i)–(iii), not an unconditional proof of MS2.

Proof. Set

$$R_\lambda := (A - u_0)k_\lambda = \mu_\lambda \tilde{u} + h_\lambda.$$

By the two residual lemmas,

$$\|R_\lambda\| \le |\mu_\lambda| + \|h_\lambda\| \le s_\lambda + p_\lambda.$$

Now decompose $k_\lambda = k_{\mathrm{cl}} + k_\perp$ with $k_{\mathrm{cl}} \in V_\lambda$ and $k_\perp \in V_\lambda^\perp$. Since V_λ is an A-invariant spectral subspace,

$$\langle k_\perp, (A - u_0)k_{\mathrm{cl}} \rangle = 0,$$

and therefore

$$g_* \|k_\perp\|^2 \le \langle k_\perp, (A - u_0)k_\perp \rangle = \langle k_\perp, R_\lambda \rangle \le \|k_\perp\| \, \|R_\lambda\|.$$

Dividing by $\|k_\perp\|$ gives

$$\|k_\perp\| \le \frac{\|R_\lambda\|}{g_*} \le \frac{s_\lambda + p_\lambda}{g_*},$$

which is exactly (23). □

7.5 Numerical calibration

Table 6: Spectral-microcluster closure: numerical calibration of the three endgame constants.

λ	$\lvert\mu_\lambda\rvert$	$\|h_\lambda\|$	g_*	$(\lvert\mu_\lambda\rvert + \|h_\lambda\|)/g_*$	$\|(I - P_{V_\lambda})k_\lambda\|$
30	2.7×10^{-7}	2.67×10^{-6}	5.05	5.83×10^{-7}	4.41×10^{-7}
40	3.0×10^{-7}	2.96×10^{-6}	5.89	5.53×10^{-7}	4.16×10^{-7}
50	2.8×10^{-7}	2.86×10^{-6}	5.90	5.32×10^{-7}	4.09×10^{-7}
75	3.1×10^{-7}	3.03×10^{-6}	7.13	4.68×10^{-7}	3.93×10^{-7}
100	2.9×10^{-7}	2.95×10^{-6}	6.82	4.75×10^{-7}	3.98×10^{-7}

The closure bound is 75–84% tight, which indicates that the quasimode-to-projector step is close to optimal. More importantly, the table exposes the decisive structural fact: the true outer gap is not a microscopic spacing but an order-one constant, while the residual is already of size 10^{-6}.

7.6 Relationship to earlier routes

The Abel–Stieltjes analysis of Section 5 remains valuable as a mechanism finder: it reveals why triangle inequalities fail and why the off-cluster cancellation is highly structured. But the final closure no longer depends on the tail-vs-gap conjecture. The Three-Lemma route is strictly sharper: it replaces one monolithic conjecture by a spectral microcluster statement and two residual estimates, each directly visible in the operator decomposition.

8 Conditional Reduction to the Riemann Hypothesis

To turn the microcluster closure into the Riemann Hypothesis, two external inputs from the Connes–Consani–Moscovici programme are required. Both are explicitly formulated by their authors as proof *strategies*, not as completed theorems: [1] describes itself as "strategy toward a proof of the Riemann Hypothesis" and [8] as "outlining a potential proof strategy based on establishing convergence of zeros from finite to infinite Euler products". We therefore record them here as named conjectural inputs and present Theorem 8.3 below as a *conditional reduction*, not as a proof of RH.

Conjecture 8.1 (CCM truncation faithfulness, after [1, 8])**.** *Assume MS2 holds along an admissible truncation regime. Then the corresponding even CCM approximants satisfy the real-zero property for the truncated* ξ_λ*.*

Theorem 8.2 (Hurwitz passage, after Fact 6.4 of [8])**.** *If the CCM approximants* ξ_λ *have the real-zero property along a sequence* $\lambda_n \to \infty$*, then the Fourier transforms* $\widehat{\xi_{\lambda_n}}$ *converge to the Riemann* Ξ*-function uniformly on closed substrips* $|\Im(z)| \le \delta < 1/2$ *at rate* $O(\lambda^{-1/2-\delta})$*, and by Hurwitz' theorem on zeros of locally uniform limits of entire functions of order one, the real-zero property passes to* Ξ*. Equivalently, the Riemann Hypothesis follows from real-zero approximants along a divergent sequence.*

Status of this statement. As stated by Connes in [8, Fact 6.4 of §6.5], the Fourier convergence with the stated rate is presented as an established consequence of the classical Slepian–Pollak prolate spheroidal wave function theory [8, Theorem 1 of reference 47], not as a conjecture. The "strategy" framing in the abstract of [8] refers to the overall RH-architecture (the combination of Fact 6.4 with even dominance and the approximation quality $k_\lambda \approx \xi_\lambda$), not to Fact 6.4 itself. The remaining open analytical questions identified by Connes in [8, §6.6] are: (i) simple even ground state of QW_λ (even dominance), addressed in the companion paper [11]; (ii) approximation quality of the educated guess k_λ versus the true ground state ξ_λ, which is precisely the spectral microcluster question MS2 addressed in Section 7 of the present paper.

Theorem 8.3 (Conditional reduction of RH via microcluster closure)**.** *Assume Theorem 8.1, Theorem 8.2, and the three endgame statements of Theorem 7.6. Then the Riemann Hypothesis holds.*

Proof. Theorem 7.6 gives MS2 from the three endgame statements (scalar secular cancellation, projected Poisson control, coercive complement). By Theorem 8.1, MS2 forces the real-zero property for the CCM approximants. By Theorem 8.2, this real-zero property passes to the completed zeta function in the limit. Therefore the Riemann Hypothesis holds. □

Honest statement of conditioning. The above is *not* a proof of the Riemann Hypothesis. After the clarifications above, the conditioning consists of:

- *External established input:* the Slepian–Pollak prolate wave function convergence theorem ([8, Fact 6.4 / Theorem 1 of Ref. 47]), which gives Theorem 8.2 as an analytical consequence of classical prolate-wave theory; and the Connes–van Suijlekom quadratic-form real-zero criterion [9], fully proved.

- *Internal open inputs:* the three endgame statements of Section 7 (scalar secular cancellation, projected Poisson quasimode, coercive complement), which together close the spectral microcluster mechanism for MS2. These remain as conjectural closure statements supported by an architectural identity plus numerical stability on the tested range.

- *Internal open input (Connes' own remaining step):* simple even ground state of QW_λ (Connes 2026 §6.6 item (i)), addressed in the companion paper [11].

The present paper addresses the MS2 question; the Even Dominance question is handled in [11]. Both papers together attempt to discharge exactly the two "remaining steps" that Connes identifies in [8, §6.6].

Remark 8.4 (Independent route). A separate conditional reduction of the Riemann Hypothesis via *even dominance* of the Weil quadratic form—using spectral positivity methods rather than the microcluster mechanism—is presented in [11], where it is itself conditional on a quantitative variational gap conjecture for the odd-sector minimum eigenvalue plus the same Connes program. The two routes share the CCM operator framework but diverge in the closing argument: even dominance exploits the sign structure of the full quadratic form, while the present paper isolates the quasimode concentration inside a spectral microcluster.

9 Conclusion: The Zookeeper's Legacy

Connes' spectral programme, culminating in the CCM Fourier basis [1], provides the first computationally accessible framework in which the low-energy mechanism behind the zeta zeros can be tested mode by mode and eigenvalue by eigenvalue with arbitrary precision.

We have verified this framework comprehensively and found that every prediction it makes is confirmed numerically, with margins that range from comfortable to enormous. The 15-digit cancellation at Riemann zeros, the resolvent identity $\alpha = R$, the Poisson kernel's near-perfect cluster localisation, the sign coherence of the spectral multiplier, the $\eta \approx 1$ distribution—all of these are consequences of a single structural fact: the rank-one pole term and the arithmetic perturbation are locked in a rigid algebraic embrace.

What this paper proves analytically:

(i) The *resolvent identity* (Theorem 3.1): Riemann zeros are resonances $\alpha(\gamma) = R(\gamma, w_{\min})$.

(ii) The η*-parametrisation* (Theorem 4.2): the local modewise inequality is equivalent to $\eta_j \in (0, 2)$ for all regular modes.

(iii) The *lower bound* $\eta_j > 0$ (Theorem 4.4), via Cauchy interlacing.

(iv) The *Three-Lemma microcluster closure* (Theorem 7.6), which turns scalar secular cancellation, projected Poisson control, and the outer spectral gap into an explicit MS2 bound.

(v) The *Conditional reduction of RH inside the same paper* (Theorem 8.3), obtained by combining the microcluster closure with the two conjectural CCM external inputs (truncation faithfulness, Hurwitz passage); this is a reduction, not a proof of RH.

What this paper verifies numerically:

(i) The upper bound $\eta_j < 2$ with margin ≥ 0.99 (all 117+ modes).

(ii) The Abel–Stieltjes Tail-vs-Gap bound < 1 (all 117 modes).

(iii) The endgame constants s_λ, p_λ, and g_* for $\lambda = 3, \dots, 100$ (N up to 120), with the closure bound 75–84% tight.

What remains external to the internal closure argument:

(O1) **Truncation faithfulness and limit passage:** the present paper works inside the finite-dimensional CCM Fourier model. To pass from the model to the full Weil positivity statement one still needs the external CCM transfer from the Fourier-Galerkin regime to the $\lambda \to \infty$ limit.

(O2) **MS1 / simple-even input:** the simplicity and even-sector properties used in the broader CCM programme are not reproved here.

The zookeeper built the enclosure. We have now isolated the resonant cage, replaced the old ad hoc fence by a spectral one, shown how the Poisson kernel is forced into it by a three-lemma argument, and included the final RH deduction in the same manuscript.

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Data availability

All computational scripts and numerical logs are available at https://github.com/research-line/functional-stability-theory/tree/main/masters/zookeeper.

AI disclosure

This work involved substantial use of AI assistants (Anthropic Claude and OpenAI ChatGPT) for numerical computation, algebraic verification, proof strategy development, and manuscript preparation. The AI contributions are documented in detail in the supplementary materials. All mathematical claims have been independently verified by the author.

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